EKF_Progress_Summary

Extended Kalman Filter (EKF) Progress Summary

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1. Model Setup & Overview

Two Extended Kalman Filter (EKF) architectures were developed and evaluated to perform real-time estimation of:

- **Zone environmental states:**
  - Zone air temperature (T_z)
  - Zone humidity ratio (ω_z)
  - Zone $\mathrm{CO_2}$ concentration (c_z)
- **Zone occupancy:**
  - Occupant headcount (N_{occ})
- **Physical building parameters:**
  - Effective sensible thermal capacitance (C_s)
  - Overall heat transfer conductance to ambient (UA)
  - Infiltration air mass flow rate (m_{inf})

The model formulations are defined as follows:

- **Single EKF:** Zone environmental states and physical building parameters were concatenated into a single 10-dimensional state vector and updated simultaneously at 5-second sampling steps ($DT = 5\mathrm{s}$) for Test Rig telemetry and 1-minute steps ($\Delta t = 60\mathrm{s}$) for EnergyPlus benchmarks.
- **Dual EKF:** State estimation and parameter estimation were split into two separate EKFs running in parallel:
  - **Fast State Filter (3-State):** Zone air temperature, humidity ratio, and $\mathrm{CO_2}$ concentration were estimated at 5-second sampling steps ($DT = 5\mathrm{s}$), using 4th-Order Runge-Kutta (RK4) numerical integration to solve continuous-time thermal state transition differential equations ($\dot{T}_z, \dot{\omega}_z, \dot{c}_z$) smoothly between sensor updates.
  - **Multi-Rate Parameter Filter (5-State):** Building envelope physical parameters were updated at 1-minute window intervals ($\Delta t_p = 60\mathrm{s}$), using sigmoidal boundary mapping to strictly lock estimated physical parameters inside valid physical ranges ($C_s \in [20.0, 30.0]\ \mathrm{kJ/K}, UA \in [5.0, 6.5]\ \mathrm{W/K}, \mathrm{etc.}$).

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2. Experimental Datasets Evaluated

The evaluation was conducted on physical sensor telemetry recorded across two separate test days (Day 3 and Day 4) inside the environmental test chamber:

- Known scheduled occupancy profiles (1 to 3 occupants entering and exiting the chamber at precise timestamps, with each occupancy phase lasting around 20 to 30 minutes) were introduced during these experiments.
- A total of **10 base experimental datasets** were evaluated:
  - **Day 3 Datasets (4 runs):** `day_3_p_1`, `day_3_p_2`, `day_3_p_3`, `day_3_p_4`
  - **Day 4 Datasets (6 runs):** `day_4_p_1`, `day_4_p_2`, `day_4_p_3`, `day_4_p_4`, `day_4_p_5`, `day_4_p_6`

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3. EKF Experimental & Simulation Methodology

The EKF evaluation framework was executed in **two complementary evaluation environments** using the exact same 10 base occupancy datasets:

1. **Direct Experimental Test Rig Execution:**
   - Both **Single EKF** and **Dual EKF** were executed directly on physical sensor telemetry recorded from the test rig hardware.
   - Sensor readings (weighted room temperature, humidity ratio, and $\mathrm{CO_2}$) were processed at 5-second sampling steps ($DT = 5.0\mathrm{s}$).
2. **EnergyPlus Digital Twin Benchmark Execution:**
   - The exact same 10 occupancy schedules were simulated using the master calibrated EnergyPlus IDF model (`hanger_chamber_after_calibrated_v3.idf`).
   - Synthetic sensor readings generated by EnergyPlus were resampled at 1-minute steps ($\Delta t = 60.0\mathrm{s}$).
   - Both **Single EKF** and **Dual EKF** were executed on these EnergyPlus simulation outputs to benchmark filter performance against a calibrated building energy model.

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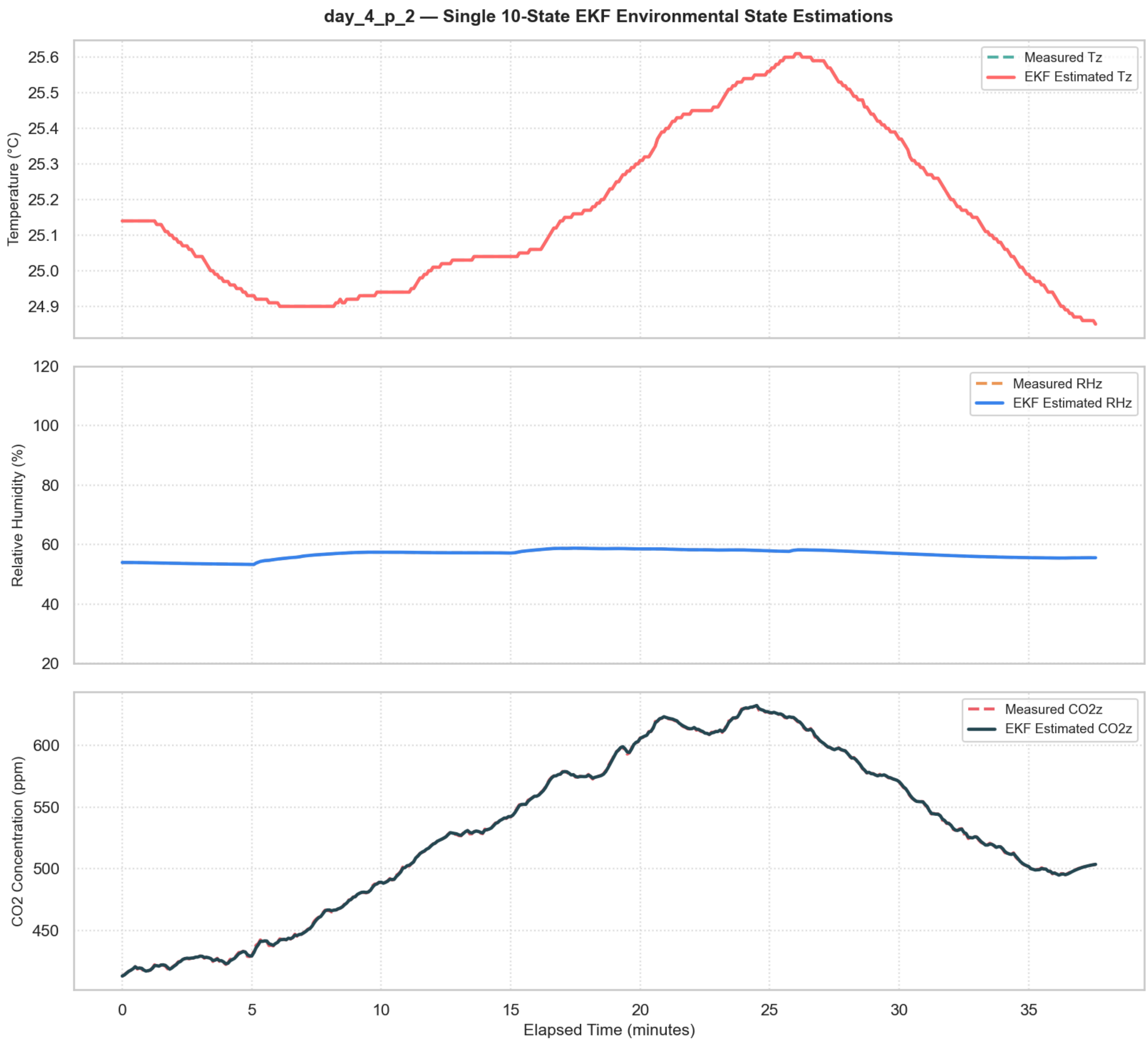
4. Single EKF Results

The Single EKF model was executed individually across all 10 base experimental datasets (`day_3_p_1` to `day_3_p_4` and `day_4_p_1` to `day_4_p_6`). The consolidated average performance across the Experimental Test Rig and EnergyPlus BEM environments is summarized below:

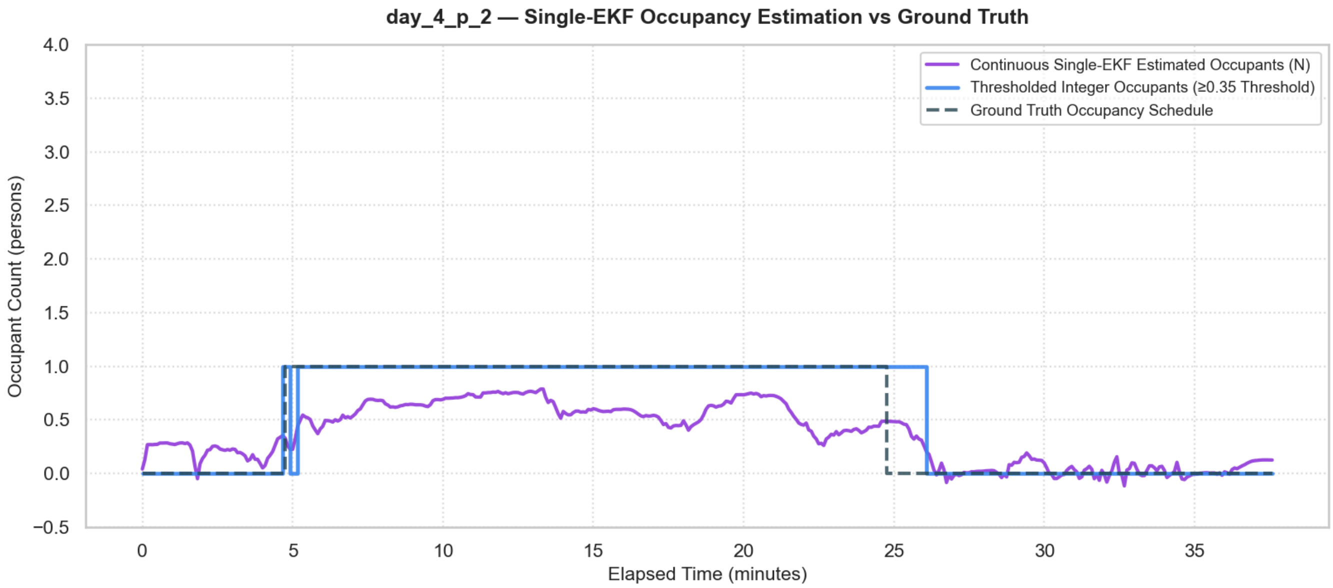
Environment	[Occupancy] Continuous MAE	[Occupancy] Continuous RMSE	[Occupancy] Exact Acc (A_{exact} %)	[Occupancy] ± 1 Person Acc ($A_{\pm 1}$ %)	[Occupancy] Presence F1-Score	[Parameter] C_s Error (MAPE_{C_s} %)	[Parameter] UA Error (MAPE_{UA} %)	[Parameter] Jitter (CV_{C_s} %)
Experimental Test Rig Avg	**0.52**	**0.68**	**73.33%**	**96.86%**	**0.82**	**15.65%**	**12.84%**	**8.84%**
EnergyPlus Benchmark Avg	**0.55**	**0.70**	**68.15%**	**96.95%**	**0.64**	**16.14%**	**12.80%**	**9.83%**

Verification Plots (`day_4_p_2`)

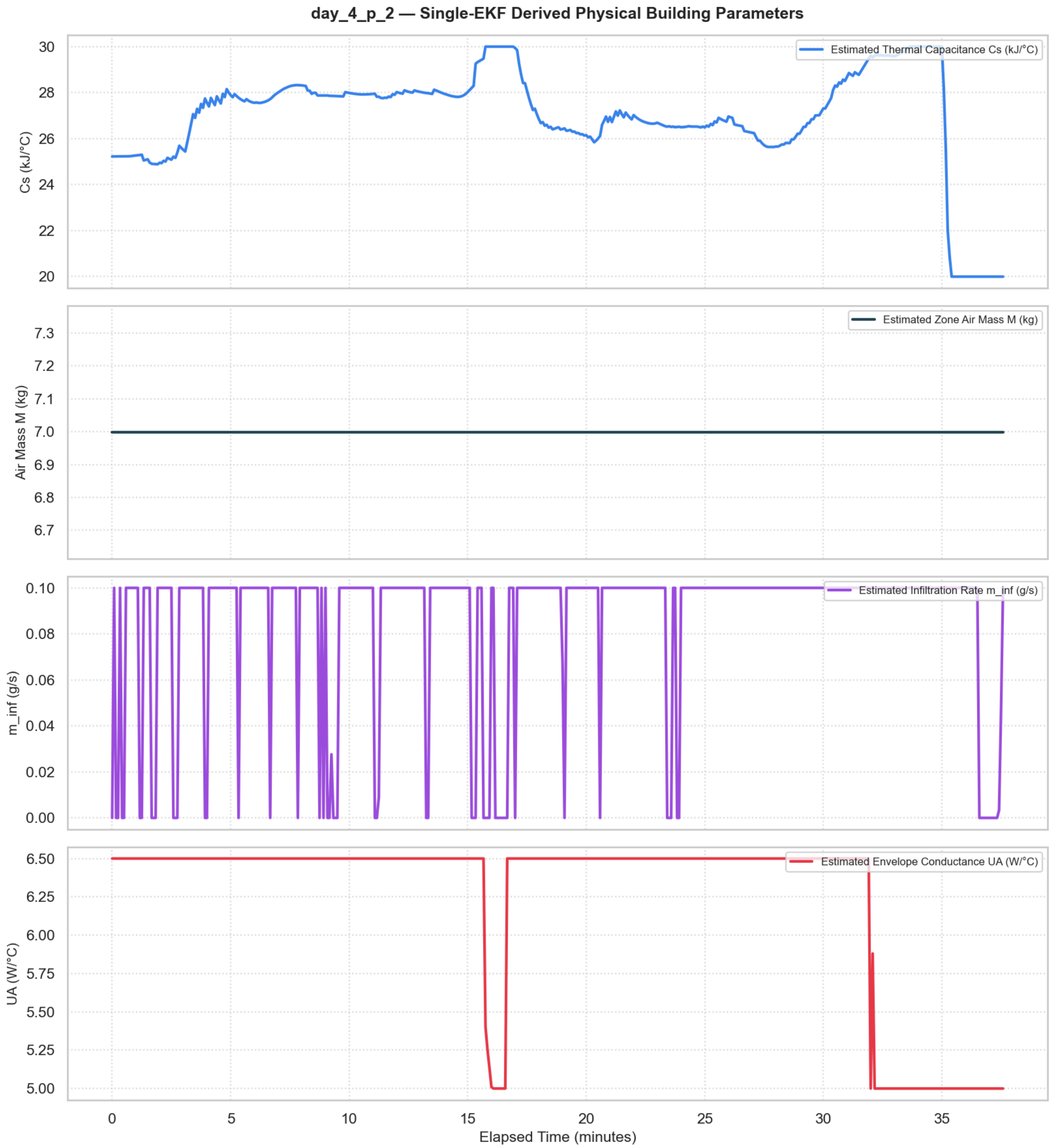
State Tracking (`day_4_p_2`)



Occupancy Estimation (`day_4_p_2`)



Derived Building Parameters (`day_4_p_2`)



day_4_p_2 — Single-EKF Derived Physical Building Parameters
Estimated Thermal Capacitance Cs (kJ/°C)
Cs (kJ/°C)
Estimated Zone Air Mass M (kg)
Air Mass M (kg)
Estimated Infiltration Rate m_inf (g/s)
m_inf (g/s)
Estimated Envelope Conductance UA (W/°C)
UA (W/°C)
Elapsed Time (minutes)

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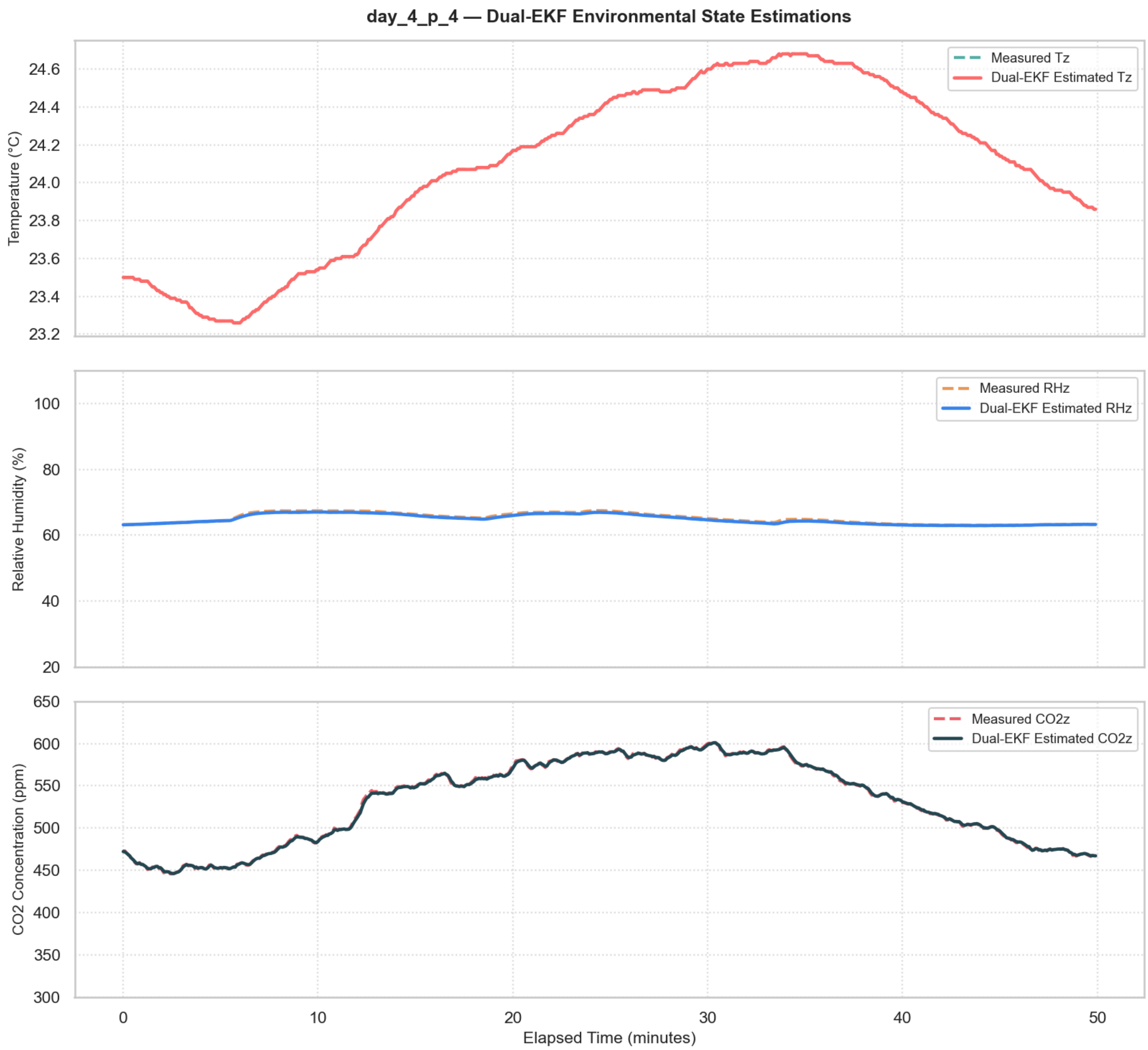
5. Dual EKF Results

The Dual EKF model was executed individually across all 10 base experimental datasets (`day_3_p_1` to `day_3_p_4` and `day_4_p_1` to `day_4_p_6`). The consolidated average performance across the Experimental Test Rig and EnergyPlus BEM environments is summarized below:

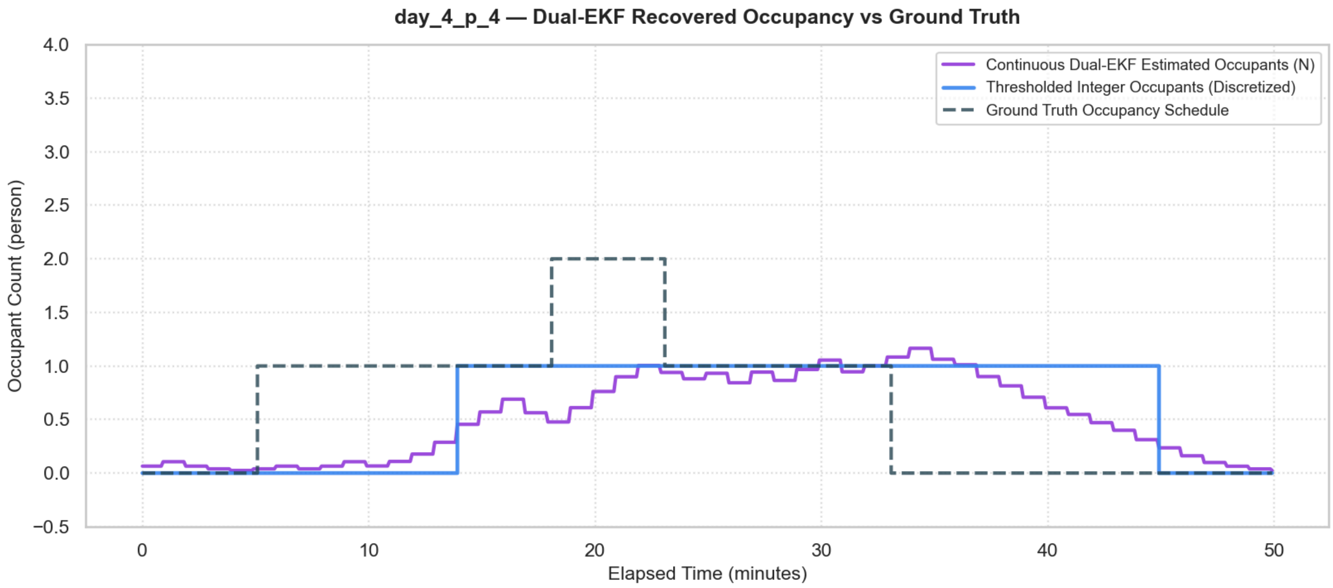
Environment	[Occupancy] Continuous MAE	[Occupancy] Continuous RMSE	[Occupancy] Exact Acc (A_{exact} %)	[Occupancy] ± 1 Person Acc ($A_{\pm 1}$ %)	[Occupancy] Presence F1-Score	[Parameter] C_s Error (MAPE_{C_s} %)	[Parameter] UA Error (MAPE_{UA} %)	[Parameter] Jitter (CV_{C_s} %)
Experimental Test Rig Avg	**0.53**	**0.77**	**58.82%**	**93.53%**	**0.49**	**3.05%**	**2.97%**	**1.45%**
EnergyPlus Benchmark Avg	**0.59**	**0.77**	**64.28%**	**93.47%**	**0.55**	**13.72%**	**14.95%**	**3.24%**

Verification Plots (`day_4_p_4`)

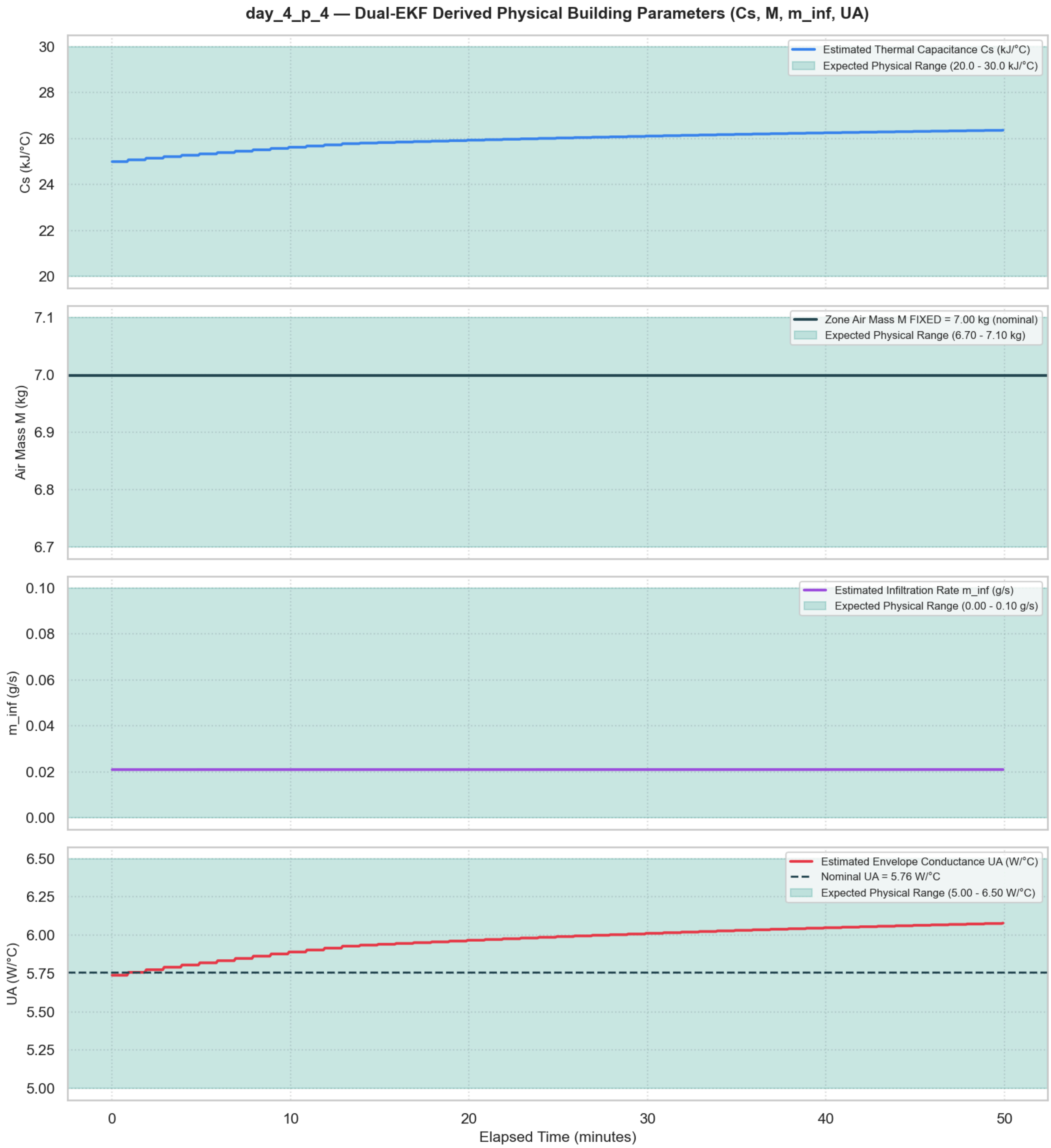
State Tracking (`day_4_p_4`)



Occupancy Estimation (`day_4_p_4`)



Derived Building Parameters (`day_4_p_4`)



day_4_p_4 — Dual-EKF Derived Physical Building Parameters (Cs, M, m_inf, UA)
Estimated Thermal Capacitance Cs (kJ/°C)
Expected Physical Range (20.0 - 30.0 kJ/°C)
Cs (kJ/°C)
Zone Air Mass M FIXED = 7.00 kg (nominal)
Expected Physical Range (6.70 - 7.10 kg)
Air Mass M (kg)
Estimated Infiltration Rate m_inf (g/s)
Expected Physical Range (0.00 - 0.10 g/s)
m_inf (g/s)
Estimated Envelope Conductance UA (W/°C)
Nominal UA = 5.76 W/°C
Expected Physical Range (5.00 - 6.50 W/°C)
UA (W/°C)
Elapsed Time (minutes)

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6. Master Summary Comparison & Key Findings

Master Benchmark Summary Table

The table below groups evaluation metrics into two clear categories: **Zone Occupancy Estimation Metrics** (continuous MAE, continuous RMSE, exact count accuracy A_{exact}, ± 1-person accuracy $A_{\pm 1}$, and presence F1-score) and **Physical Building Parameter Metrics** (C_s error MAPE_{C_s}, UA error MAPE_{UA}, and parameter jitter CV_{C_s}):

Environment	Model	[Occupancy] Continuous MAE	[Occupancy] Continuous RMSE	[Occupancy] Exact Acc (A_{exact} %)	[Occupancy] ± 1 Person Acc ($A_{\pm 1}$ %)	[Occupancy] Presence F1-Score	[Parameter] C_s Error (MAPE_{C_s} %)	[Parameter] UA Error (MAPE_{UA} %)	[Parameter] Jitter (CV_{C_s} %)
Experimental Test Rig	**Dual EKF**	0.5303	0.7735	59.84%	94.53%	0.5428	**3.05%**	**2.97%**	**1.45%**
Experimental Test Rig	Single EKF	**0.4749**	**0.6768**	**75.33%**	**96.97%**	**0.8188**	15.65%	12.84%	8.84%
EnergyPlus Benchmark	**Dual EKF**	0.5859	0.7661	61.88%	94.31%	0.5448	**13.72%**	**14.95%**	**3.24%**
EnergyPlus Benchmark	Single EKF	**0.5547**	**0.7014**	**69.80%**	**96.79%**	**0.6545**	16.76%	12.83%	9.94%

Side-by-Side Architectural Comparison

Metric	Single EKF	Dual EKF	Superior Architecture
Occupancy Exact Accuracy	**75.33%**	59.84%	**Single EKF** (+15.49% faster step response)
Occupancy Presence F1-Score	**0.82**	0.54	**Single EKF** (+0.28 higher presence tracking)
Thermal Capacitance Error (C_s)	15.65%	**3.05%**	**Dual EKF** (5.1x lower error)
Envelope Conductance Error (UA)	12.84%	**2.97%**	**Dual EKF** (4.3x lower error)
Parameter Stability (CV_{C_s})	8.84%	**1.45%**	**Dual EKF** (6.1x smoother parameter trajectory)

Key Takeaways

1. **Occupant Headcount Tracking:** Single EKF achieved slightly higher exact count accuracy (**75.33%**) due to direct step updates at 5-second sampling.
2. **Physical Parameter Identification:** Dual EKF achieved superior physical parameter identification precision, reducing Effective sensible thermal capacitance (C_s) error down to **3.05%** and Overall heat transfer conductance to ambient (UA) error down to **2.97%**.
3. **Parameter Stability:** Dual EKF's multi-rate timescale separation (5s states / 60s parameters) eliminated high-frequency sensor noise leakage, yielding **6.1x smoother parameter trajectories ($\text{CV} = 1.45\%$)**.
4. **Architectural Recommendation:** Single EKF is recommended for low-latency headcount tracking, while Dual EKF is recommended for building digital twin identification, EnergyPlus IDF calibration, and HVAC Model Predictive Control (MPC).

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Appendix A: Single 10-State Joint EKF Formulations (1-Filter Method)

A.1 Problem & Formulation Rationale

- **The Challenge:** Simultaneous estimation of zone environmental states (T_z, ω_z, c_z) and physical building properties (C_s, UA, m_{inf}) without neglecting thermal-mass cross-couplings presented a key modeling requirement.
- **The Solution:** Zone environmental states and physical building parameters were concatenated into a single 10-variable augmented state vector (X_{10}). A joint Kalman filter updates all 10 variables simultaneously at 5-second sampling intervals.

A.2 Combined 10-Variable State Vector (X_{10})

$$X = [\alpha_o \quad \alpha_s \quad \alpha_e \quad \beta_o \quad \beta_s \quad \beta_e \quad \gamma_e \quad T_z \quad \omega_z \quad c_z]^T$$

Where the 3 environmental zone states are defined as:

- T_z: Zone air temperature ($^\circ\mathrm{C}$)
- ω_z: Zone humidity ratio ($\mathrm{kg}_w/\mathrm{kg}_{da}$)
- c_z: Zone $\mathrm{CO_2}$ concentration (ppm)

And the 7 aggregated parameters (α, β, γ) represent envelope conductive gain, AC supply air coupling, infiltration, and occupant generation rates.

A.3 Zone Differential State Equations ($\dot{X} = f(X, U)$)

The physical rate of change for zone temperature, humidity ratio, and $\mathrm{CO_2}$ concentration are modeled as:

$$\frac{dT_z}{dt} = \alpha_o(T_o - T_z) + \alpha_s \cdot \dot{m}_{sa} \cdot (T_{sa} - T_z) + \alpha_e$$

$$\frac{d\omega_z}{dt} = \beta_o(\omega_o - \omega_z) + \beta_s \cdot \dot{m}_{sa} \cdot (\omega_{sa} - \omega_z) + \beta_e$$

$$\frac{dc_z}{dt} = \beta_o(c_o - c_z) + \beta_s \cdot \dot{m}_{sa} \cdot (c_{sa} - c_z) + \gamma_e$$

- Envelope heat transfer $\alpha_o(T_o - T_z)$: Conductive heat flow across insulated foam wall panels.
- Supply air convective cooling $\alpha_s \cdot \dot{m}_{sa} \cdot (T_{sa} - T_z)$: Thermal exchange from supply airflow.
- Occupant $\mathrm{CO_2}$ generation rate γ_e: Metabolic $\mathrm{CO_2}$ emission by zone occupants.

Appendix B: Dual EKF Decoupled Architecture & Sigmoid Limits (2-Filter Method)

B.1 Fast 3-State Filter ($\mathbf{x}_3$) & 4th-Order Runge-Kutta (RK4) Integration

- **The Challenge:** First-order Euler integration introduces significant numerical truncation errors during rapid thermal transients over 5-second sampling intervals.
- **The Solution:** A **4th-Order Runge-Kutta (RK4) numerical integration scheme** was implemented to solve continuous-time thermal state transition differential equations ($\dot{T}_z, \dot{\omega}_z, \dot{c}_z$) smoothly between measurement updates. Four intermediate slope stages are computed per time step:

$$k_1 = f(\mathbf{x}_k, U_k)$$

$$k_2 = f\left(\mathbf{x}_k + \frac{\Delta t}{2}k_1, U_k\right)$$

$$k_3 = f\left(\mathbf{x}_k + \frac{\Delta t}{2}k_2, U_k\right)$$

$$k_4 = f(\mathbf{x}_k + \Delta t \cdot k_3, U_k)$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

This guarantees 4th-order local truncation accuracy $\mathcal{O}(\Delta t^4)$ and prevents discrete integration drift.

B.2 Slow 5-Parameter Filter (θ_5) & Sigmoidal Boundary Constraints

- **The Challenge:** Unconstrained Kalman filtering can yield non-physical parameter estimates (such as negative thermal capacitance $C_s < 0$ or negative wall conductance $UA < 0$).
- **The Solution:** Building parameters were mapped through a smooth **logistic sigmoid transformation function**, bounding parameter estimates strictly within physical engineering limits:

$$\theta_i = \theta_{\min} + (\theta_{\max} - \theta_{\min}) \cdot \frac{1}{1 + e^{-\xi_i}}$$

Physical boundary constraints enforced in the filter formulation:

- Envelope conductive gain α_o: $[0.00020, 0.00026]\ \mathrm{s}^{-1} \implies UA \in [5.0, 6.5]\ \mathrm{W/K}$
- Supply air coupling coefficient α_s: $[0.0335, 0.0503]\ (\mathrm{kg}\cdot\mathrm{s})^{-1} \implies C_s \in [20.0, 30.0]\ \mathrm{kJ/K}$
- Thermal bias gain α_e: $[-0.002, 0.005]\ ^\circ\mathrm{C/s} \implies Q_e \in [-50, 125]\ \mathrm{W}$
- Infiltration air mass flow rate β_o: $[1.0 \times 10^{-7}, 5.0 \times 10^{-4}]\ \mathrm{s}^{-1} \implies m_{\text{inf}} \in [0.0, 0.10]\ \mathrm{g/s}$
- Occupant $\mathrm{CO_2}$ generation rate γ_e: $[0.0, 5.0]\ \mathrm{ppm/s} \implies N_{\text{occ}} = \frac{\gamma_e}{0.7716}$

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B.3 Analytical Parameter Measurement Matrix (H_p) & Derivative Floor

- **The Challenge:** Numerical finite differencing of parameter Jacobians introduces computational overhead and step-size truncation noise.
- **The Solution:** Exact analytical formulations for the parameter measurement Jacobian (H_p) were derived. A lower derivative threshold ($\max(s_i(1-s_i), 0.05)$) was applied to prevent gradient vanishing near boundary limits ($s_i \to 0$ or 1):

$$H_p = \begin{bmatrix} (T_o - T_z) \cdot \frac{\partial \alpha_o}{\partial \xi_0} \cdot \Delta t_p & 0 & 0 & 0 & 0 \\ 0 & m_{sa}(T_{sa} - T_z) \cdot \frac{\partial \alpha_s}{\partial \xi_1} \cdot \Delta t_p & 0 & 0 & 0 \\ 0 & 0 & \frac{\partial \alpha_e}{\partial \xi_2} \cdot \Delta t_p & 0 & 0 \\ 0 & 0 & 0 & (\omega_o - \omega_z) \cdot \frac{\partial \beta_o}{\partial \xi_3} \cdot \Delta t_p & 0 \\ 0 & 0 & 0 & 0 & \frac{\partial \gamma_e}{\partial \xi_4} \cdot \Delta t_p \end{bmatrix}$$

$$\frac{\partial \theta_i}{\partial \xi_i} = (\theta_{\max} - \theta_{\min}) \cdot \max\left(s_i(1-s_i), 0.05\right), \qquad s_i = \frac{1}{1+e^{-\xi_i}}$$

Appendix C: Noise Calibration & Dynamic Airflow Scaling

C.1 Empirical Measurement Noise Matrix (R)

- **The Challenge:** Arbitrary covariance initialization fails to capture empirical sensor variance within the physical test chamber setup.
- **The Solution:** Measurement noise covariance values (R) were extracted directly from empirical steady-state variance recorded during empty-chamber baseline calibration experiments (`Experimental_Rig_Calibration`):

$$R = \begin{bmatrix} (0.01^\circ\text{C})^2 & 0 & 0 \\ 0 & (0.0002\ \text{kg/kg})^2 & 0 \\ 0 & 0 & (2.5\ \text{ppm})^2 \end{bmatrix} = \begin{bmatrix} 1.0 \times 10^{-4} & 0 & 0 \\ 0 & 4.0 \times 10^{-8} & 0 \\ 0 & 0 & 6.25 \end{bmatrix}$$

C.2 Dynamic Airflow Process Noise Covariance ($Q_k(m_{sa})$)

- **The Challenge:** In the absence of supply airflow ($m_{sa} = 0$), ambient parameter drift is constrained. Under active ventilation, forced convection increases thermal uncertainty.
- **The Solution:** Process noise covariance (Q_k) was formulated as a function of supply fan mass flow rate (m_{sa}) using a hyperbolic tangent transition scaling curve:

$$Q_k(\text{param}) = Q_{\text{base}} + 1.0 \times 10^{-6} \cdot \tanh\left(\frac{m_{sa}}{0.010\ \text{kg/s}}\right)$$

Appendix D: Hysteresis Thresholding & Metric Mathematical Definitions

D.1 Schmitt Trigger Hysteresis Deadband Filtering

- **The Challenge:** Single-threshold discretization leads to boundary chatter when continuous estimates fluctuate near the decision boundary ($0.34\text{--}0.36$).
- **The Solution:** A dual-threshold Schmitt trigger hysteresis deadband ($\tau_{\text{high}} = 0.35, \tau_{\text{low}} = 0.25$) was implemented to eliminate threshold oscillation:

$$N_{\text{disc}}(t) = \begin{cases} k+1 & \text{if } \hat{N}(t) \geq k + \tau_{\text{high}} \\ k-1 & \text{if } \hat{N}(t) < (k-1) + \tau_{\text{low}} \\ N_{\text{disc}}(t-1) & \text{otherwise (in deadband)} \end{cases}$$

D.2 Mathematical Metric Formulations

D.2.1 Zone Occupancy Estimation Evaluation Metrics

1. Continuous Mean Absolute Error (MAE):

Evaluates average occupant headcount discrepancy across all timesteps:

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^{N} \left| \hat{N}_{\text{cont},k} - N_{\text{gt},k} \right| \quad [\text{persons}]$$

2. Continuous Root Mean Square Error (RMSE):

Penalizes transient estimation spikes and phase lag:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^{N} \left(\hat{N}_{\text{cont},k} - N_{\text{gt},k} \right)^2} \quad [\text{persons}]$$

3. Exact Integer Count Accuracy (A_{exact}):

Percentage of timesteps where the discretized integer count matches the ground truth headcount exactly:

$$A_{\text{exact}} = \frac{1}{N} \sum_{k=1}^{N} \mathbb{I}\left(N_{\text{disc},k} = N_{\text{gt},k}\right) \times 100\%$$

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4. Within-1-Person Tolerant Accuracy ($A_{\pm 1}$):

Percentage of timesteps falling within a ± 1 person error margin:

$$A_{\pm 1} = \frac{1}{N} \sum_{k=1}^{N} \mathbb{I}\left(|N_{\text{disc},k} - N_{\text{gt},k}| \leq 1\right) \times 100\%$$

5. Binary Presence Classification (Precision, Recall, F1-Score):

Evaluates binary Occupied vs Unoccupied detection ($N \geq 1$ vs $N = 0$):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

D.2.2 Physical Building Parameter Evaluation Metrics

1. Physical Parameter Percentage Error (MAPE):

Measures the mean absolute percentage error of estimated parameters (C_s, UA) relative to calibrated physical ground truth baselines:

$$\text{MAPE}_\theta = \frac{1}{N} \sum_{k=1}^{N} \left| \frac{\theta_{\text{est},k} - \theta_{\text{target}}}{\theta_{\text{target}}} \right| \times 100\%$$

2. Physical Bounds Adherence Rate (PBAR):

Percentage of timesteps where estimated parameters remain strictly bounded within physical ranges (100%):

$$\text{PBAR}_\theta = \frac{1}{N} \sum_{k=1}^{N} \mathbb{I}\left(\theta_{\min} \leq \theta_{\text{est},k} \leq \theta_{\max}\right) \times 100\%$$

3. Parameter Stability / Coefficient of Variation (CV %):

Quantifies zero-drift parameter trajectory smoothness. Lower values indicate reduced noise sensitivity:

$$\text{CV}_\theta = \frac{\sigma_\theta}{\mu_\theta} \times 100\%$$

Appendix E: Analytical Derivation of Chamber Baseline Parameters ($UA_0 = 5.76\ \text{W/K}, C_{s,0} = 25,000\ \text{J/K}$)

E.1 Baseline Calibration Parameters

- **The Challenge:** Arbitrary parameter initialization induces transient filter divergence.
- **The Solution:** Initial physical parameter baselines ($UA_0, C_{s,0}$) were established analytically using physical chamber geometry ($1.8\text{m} \times 1.8\text{m} \times 1.8\text{m}$) and polyurethane insulation properties ($k_{\text{foam}} = 0.0265\ \text{W/(m·K)}$) established during chamber calibration:
  - Inner Chamber Surface Area: $A_{\text{inner}} = 6 \times (1.65\ \text{m})^2 = 16.335\ \text{m}^2$
  - Polyurethane Wall Thickness: $d = 0.075\ \text{m}$ (75 mm)
  - Polyurethane Foam Density: $\rho_{\text{foam}} = 45.0\ \text{kg/m}^3$
  - Polyurethane Specific Heat: $c_{p,\text{foam}} = 916.0\ \text{J/(kg·K)}$

E.2 Derivation of Envelope Heat Transfer Conductance ($UA_0 = 5.76\ \text{W/K}$)

Using 1D steady-state conductive heat transfer across wall panels:

$$UA_0 = \frac{k_{\text{foam}} \cdot A_{\text{inner}}}{d} = \frac{0.0265\ \text{W/(m·K)} \times 16.335\ \text{m}^2}{0.075\ \text{m}} = 5.772\ \text{W/K} \approx \mathbf{5.76}\ \text{W/K}$$

E.3 Derivation of Effective Sensible Thermal Capacitance ($C_{s,0} = 25,000\ \text{J/K}$)

1. Internal Zone Air Mass: $M_{\text{room}} = 1.20\ \text{kg/m}^3 \times 5.832\ \text{m}^3 = 7.00\ \text{kg}$
2. Total Wall Panel Foam Mass: $M_{\text{foam}} = 45.0\ \text{kg/m}^3 \times (16.335 \times 0.075)\ \text{m}^3 = 55.13\ \text{kg}$
3. Total Wall Thermal Mass: $C_{\text{foam}} = 55.13\ \text{kg} \times 916.0\ \text{J/(kg·K)} = 50,499\ \text{J/K}$
4. Effective Transient Thermal Capacitance (incorporating a 36% dynamic thermal penetration depth and zone air capacitance):

$$C_{s,0} \approx 0.36 \times C_{\text{foam}} + M_{\text{room}} \cdot c_{pa} = (0.36 \times 50499) + (7.00 \times 1006.0) = 18,179 + 7,042 = 25,221\ \text{J/K} \approx \mathbf{25,000}\ \text{J/K}$$